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9.2.4 The Schur and Spectral Decompositions

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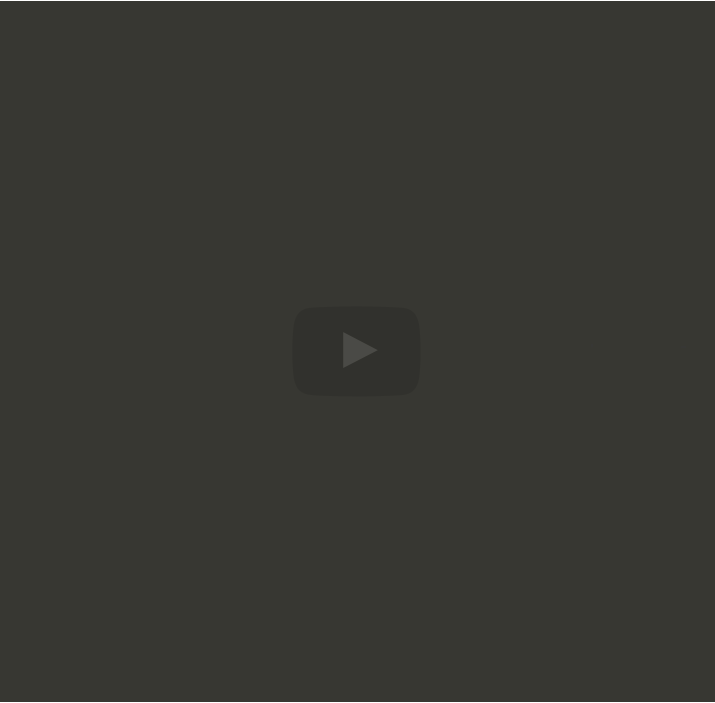
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from the new matrix.
Okay? Now to that end
we need to introduce the concept of a similarity transformation. If we have a matrix, if we have matrices A and B and Y where Y is non-singular and if B is equal to Y inverse A times Y then A and B are said to be similar. Okay? And anytime you hit a matrix with the inverse on the left and the matrix on the right, that is known as a similarity transformation. And ..

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Reading assignment

1/1 point (graded)

Read Unit 9.2.4 of the notes. [\[LINK\]](#)

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Homework 9.2.4.1

1/1 point (graded)

Let $A, B, Y \in \mathbb{C}^{m \times m}$, where Y is nonsingular, and (λ, x) an eigenpair of A

Which of the follow is an eigenpair of $B = Y^{-1}AY$:

- ☐ (λ, x)
- ☒ $(\lambda, Y^{-1}x)$
- ☐ (λ, Yx)
- ☐ $(1/\lambda, Y^{-1}x)$



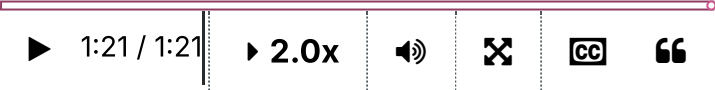
Solution

For a complete solution, see the [notes](#).

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Answers are displayed within the problem

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you cast things in terms of multiplication with general matrices and their inverses, now it depends all on the condition number of the matrix that

you're working with. So if Y is actually unitary, in other words if V can be

written as Q Hermitian transpose A times Q where Q is unitary. and this right here

is just a special case of a similarity transformation known as a unitary

similarity

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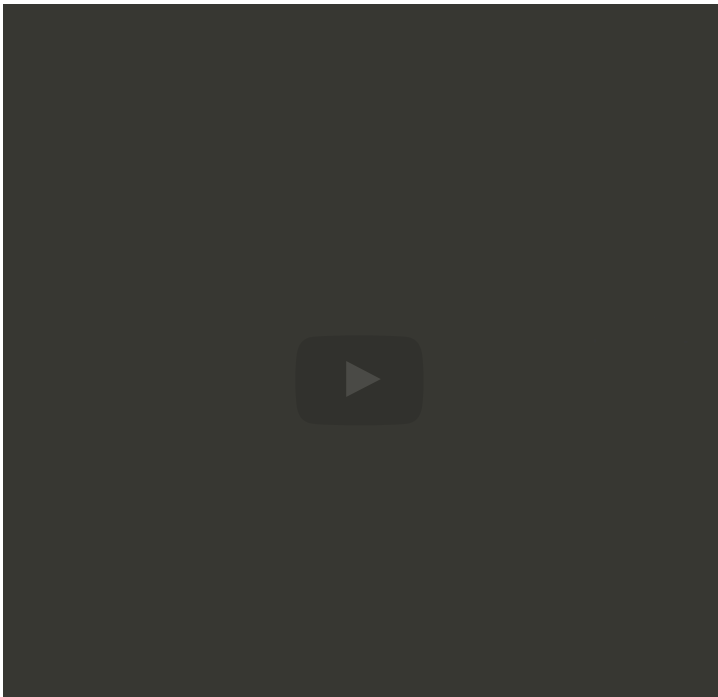
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that side. Alright? This is in the right



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Homework 9.2.4.2

1/1 point (graded)

Let $A \in \mathbb{C}^{m \times m}$, $A = QUQ^H$ be its Schur decomposition, and $X^{-1}UX = \Lambda$, where Λ is a diagonal matrix and X is nonsingular.

- How are the elements of Λ related to the elements of U ?
- How are the columns of X related to the eigenvectors of A ?

☒ done



Solution

For a complete solution, see the [notes](#).

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direction. If we can show that the result

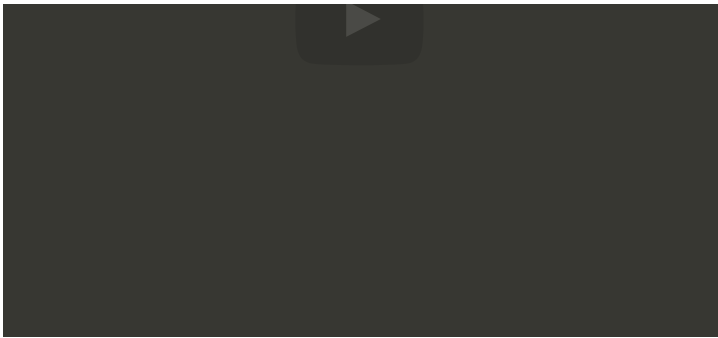
of this is upper triangular, we'd be done. But, you know, we're not that lucky yet.

We're not that good at just magically picking Q_2 . Hmm, this here is equal to

q_1 Hermitian transpose Q_2 Hermitian transpose times matrix A . And let's

actually go ahead and multiply this out and this would be A times q_1 and then A

diagonal. Actually more than that, its diagonal has to be real-valued. Okay? So that insight is known as the Spectral Decomposition Theorem. So let A -- we can replace this with the Spectral



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Decomposition Theorem. And what it says is let A in that be Hermitian. Then there exists a unitary matrix Q and diagonal Λ in $\mathbb{R}^m$ by m such that A is equal to $Q \Lambda Q^H$. And then all kinds of nice

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Homework 9.2.4.3

1/1 point (graded)
Let $A \in \mathbb{C}^{m \times m}$ be of form

$$A = \left(\begin{array}{c|c} A_{TL} & A_{TR} \\ \hline 0 & A_{BR} \end{array} \right),$$

where A_{TL} and A_{BR} are square submatrices with Schur decompositions

$$A_{TL} = Q_{TL} U_{TL} Q_{TL}^H \text{ and } A_{BR} = Q_{BR} U_{BR} Q_{BR}^H.$$

Give the Schur decomposition of A .

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Homework 9.2.4.4

1/1 point (graded)
Generalize the result in the last homework for block upper triangular matrices:

$$A = \left(\begin{array}{c|c|c|c} A_{0,0} & A_{0,1} & \cdots & A_{0,N-1} \\ \hline 0 & A_{1,1} & \cdots & A_{1,N-1} \\ \hline 0 & 0 & \ddots & \vdots \\ \hline 0 & 0 & \cdots & A_{N-1,N-1} \end{array} \right).$$

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